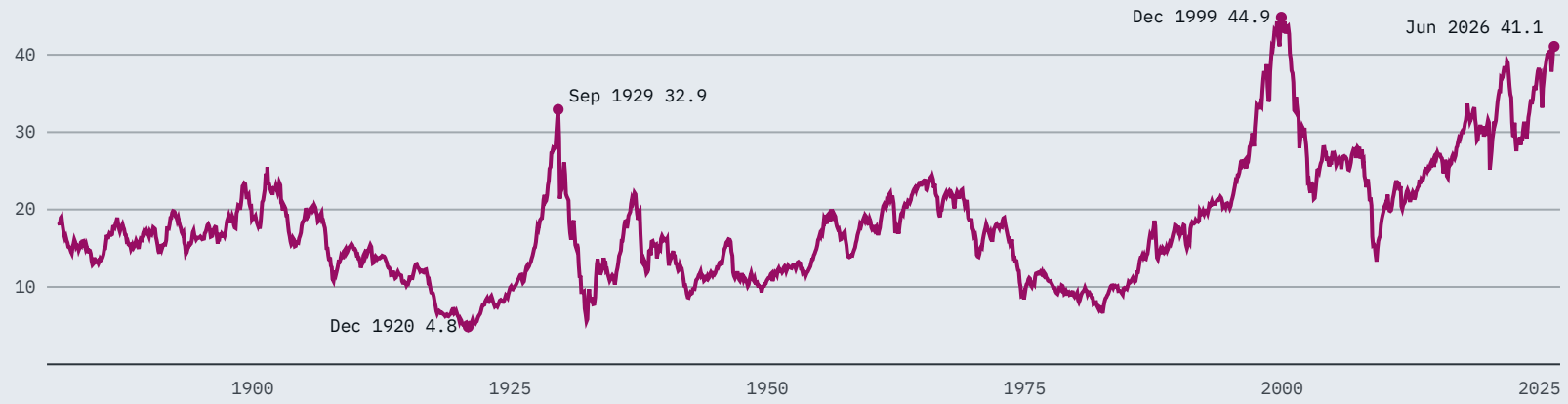
Multi-Horizon Stock Market Return Forecasting

Can 155 years of market data predict U.S. stock returns?

Raphael Erler, Richard Zallinger and Michael Prokop



CAPE: the stock price divided by the average inflation-adjusted earnings of the last 10 years. High means stocks are expensive. Monthly, 1881 to June 2026. Source: `ie_data.xls`, shillerdata.com.

Short and long horizons need different signals

Short term, 1 to 12 months

What predicts

The recent trend (momentum) and how wildly prices swing (volatility)

Price level (P/E)

Says little about the next year

Risk

Calm and wild phases come in clusters; crashes come suddenly

Long term, 5 to 10 years

What predicts

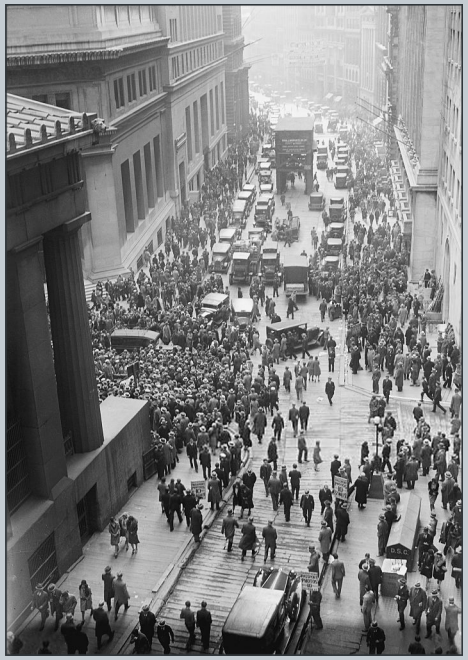
Whether stocks are cheap or expensive (CAPE, P/E)

Earnings

Rise and fall in cycles of several years, then even out

Outcome

Usually a gain, even after inflation



Outside the New York Stock Exchange after the crash, 1929. Public domain, Wikimedia Commons.

Our research question

How well can price level, recent trends, volatility and past crashes predict stock returns 1, 5 and 10 years ahead?

And does a rule-based strategy beat simply buying the index and holding it?

Five terms we use

Real return	Return after removing inflation, with dividends reinvested.
P/E and CAPE	Price divided by earnings. CAPE uses the average earnings of the last 10 years. High means expensive.
Momentum	How much the price rose or fell over the last 6 or 12 months.
Volatility	How strongly the price swings from month to month.
Drawdown	How far the price is below its previous high.



Edison's stock ticker printed prices on paper tape until the internet took over. Photo: Jaclyn Nash, public domain, Wikimedia Commons.

Three sub-questions, one per project phase

Question	Phase	What we look at	Points
How risky is each horizon?	RQ 1, exploratory analysis	How often you end with a gain after 1, 5 and 10 years, how long losses last, and why leveraged funds lose value over time.	6
Do simple rules work?	RQ 2, hypothesis tests	Does buying after a crash pay off? Do calm stocks give more return per unit of risk?	2
Can models predict returns?	RQ 3, machine learning	Predict real returns 1, 5 and 10 years ahead with linear models and tree ensembles.	10

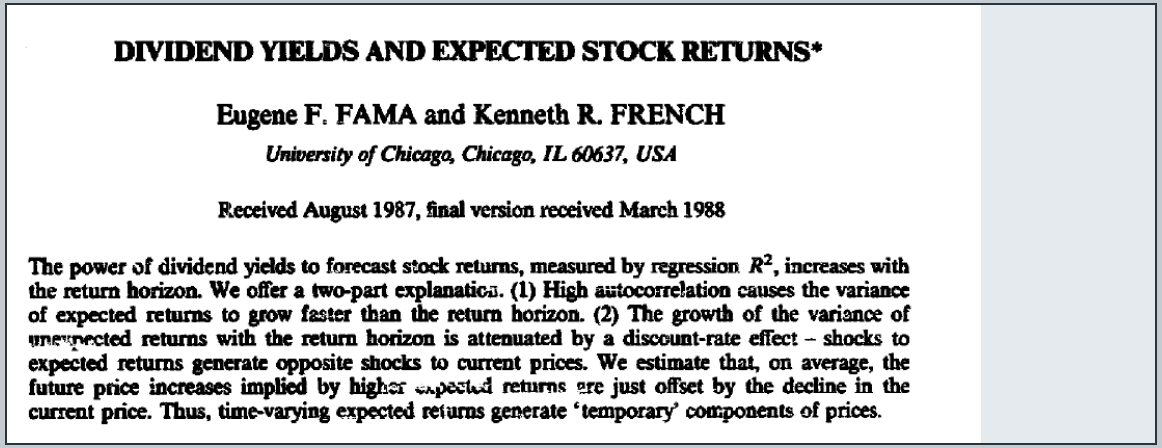
What research already shows

Work	Year	What they found
Campbell and Shiller	1988, 1998	When stocks are expensive (high CAPE), the next 10 years tend to bring lower returns.
Jegadeesh and Titman; Carhart	1993; 1997	Stocks that rose over the last 6 to 12 months tend to keep rising for a while.
Blitz and van Vliet	2007	Calm, low-volatility stocks give more return per unit of risk.
Fama and French	1988	Dividend yields predict returns, and the longer the horizon, the better.
Frazzini and Pedersen	2014	Betting against beta: low-beta assets earn more per unit of risk than high-beta ones.
Bessembinder	2018	Most single stocks do worse than the index; a few huge winners carry it.

Two papers we build on

Fama and French (1988)

The predictive power of yields grows with the horizon. This is why we forecast 1, 5 and 10 years.



DIVIDEND YIELDS AND EXPECTED STOCK RETURNS*

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Received August 1987, final version received March 1988

The power of dividend yields to forecast stock returns, measured by regression R^2, increases with the return horizon. We offer a two-part explanation. (1) High autocorrelation causes the variance of expected returns to grow faster than the return horizon. (2) The growth of the variance of unexpected returns with the return horizon is attenuated by a discount-rate effect – shocks to expected returns generate opposite shocks to current prices. We estimate that, on average, the future price increases implied by higher expected returns are just offset by the decline in the current price. Thus, time-varying expected returns generate 'temporary' components of prices.

Journal of Financial Economics 22 (1988), pages 3 to 25.

Frazzini and Pedersen (2014)

Buying low-beta and selling high-beta assets pays off in almost every market. This is the basis of our low-volatility test.

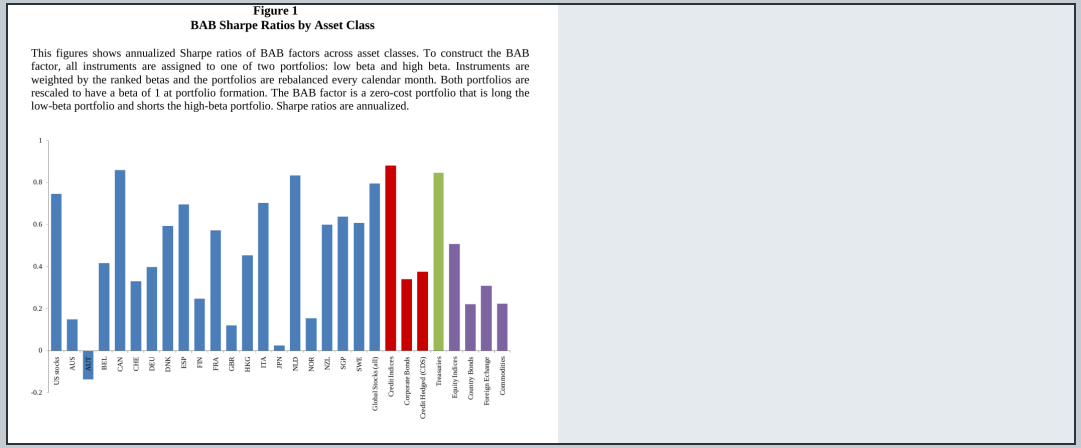
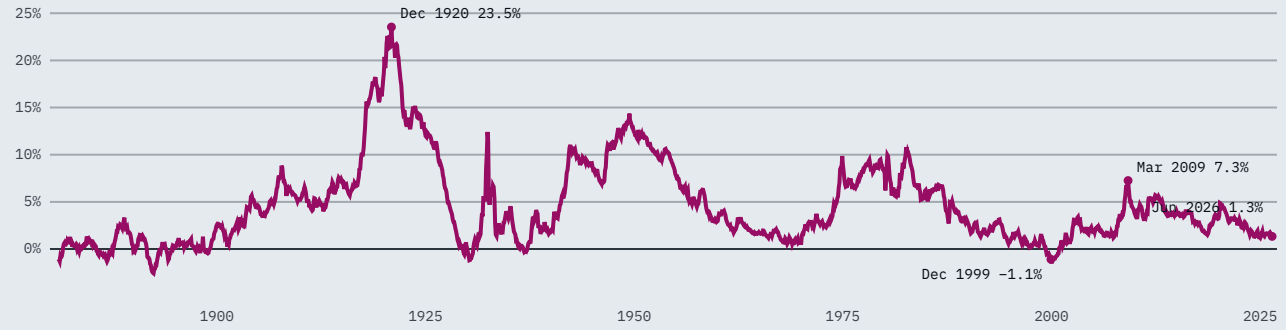


Figure 1
BAB Sharpe Ratios by Asset Class

This figures shows annualized Sharpe ratios of BAB factors across asset classes. To construct the BAB factor, all instruments are assigned to one of two portfolios: low beta and high beta. Instruments are weighted by the ranked betas and the portfolios are rebalanced every calendar month. Both portfolios are rescaled to have a beta of 1 at portfolio formation. The BAB factor is a zero-cost portfolio that is long the low-beta portfolio and shorts the high-beta portfolio. Sharpe ratios are annualized.

Figure 1, Betting Against Beta, NBER working paper 16601; Journal of Financial Economics 111 (2014), pages 1 to 25.

Bond yields help over 10 years, not over 1



Equity risk premium over time, measured as Shiller's excess CAPE yield: 1/CAPE minus the real 10-year bond yield. Monthly, 1881 to June 2026. Source: `ie_data.xls`, shillerdata.com.

Signal	Correlation with next 1-year return	with next 10-year return
Excess CAPE yield (stocks vs. bonds)	0.22	0.58
10-year bond yield alone	−0.02	0.07

Existing work: our own first check

Robert Shiller’s U.S. market data

One row per month from January 1871 to June 2026: more than 1,860 rows.

Column	Meaning	Jan 1871	Jun 2026
P	Stock index price (S&P Composite, monthly average)	4.44	7,450.03
D	Dividends paid over the year	0.26	81.70
E	Company earnings over the year	0.40	295.39
CPI	Consumer price index, to remove inflation	12.46	333.95
GS10	Interest rate of 10-year U.S. government bonds, in %	5.32	4.47

Later milestones may add daily S&P 500 prices (`yfinance`) and Kenneth French’s factor data.



Robert J. Shiller, Yale economist, Nobel Prize 2013. Photo: Presidential Office Building, Taiwan, CC BY 2.0, Wikimedia Commons.

Cleaning the data without cheating

1

Fix the dates

The file writes October 1871 as `1871.1`, the same as January. We restore `1871.10`.

2

No peeking ahead

Earnings are published about 3 months late, so each month only uses earnings from 3 months before (E_{t-3}).

3

Real total return

Remove inflation with the CPI and count dividends ($D/12$ per month) as reinvested.

4

Build the features

Compute P/E, CAPE, 6 and 12 month momentum, volatility and drawdown.

Models, baselines and a fair test

Part	Plan	Why
Linear models	Ridge, Lasso, ElasticNet	Simple and regularised, so they do not overfit noisy returns
Tree ensembles	Random Forest, XGBoost	Catch non-linear patterns, like signals that only work in some market phases
Validation	Walk-forward split with a gap	Always train on the past and test on the future; the gap keeps overlapping targets apart
Baselines	Historical average; CAPE alone	Every model must beat these. Scored with RMSE, MAE, out-of-sample R^2 and AUC

What could go wrong, and what we do about it

Overlapping targets

The 10-year returns of two neighbouring months share almost all their data. We leave gaps between training and test data.

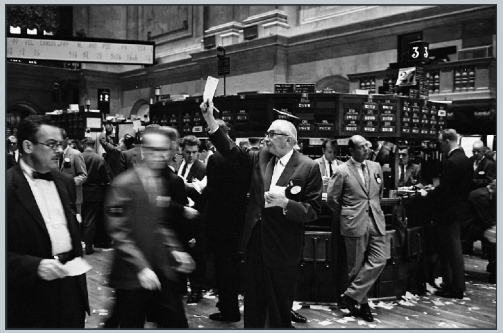
Averaged prices

Monthly average prices smooth out returns (Working's effect). We compare them with daily closing prices.

Changing markets

Markets in 1871 worked differently. We check whether the results still hold for data after 1926 alone.

Thank you. Questions and feedback welcome.



NYSE trading floor, 1963. Photo: Thomas J. O'Halloran, public domain, Wikimedia Commons.